

# Colby Dobson

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## EDUCATION

University of San Francisco, San Francisco, CA

- B.S. in Computer Science August 2021 - May 2025
  - *Minor in Mathematics*
- M.S. in Computer Science; AI Concentration August 2023 - May 2026
- President, Club Men's Volleyball August 2023 - May 2026

## WORK EXPERIENCE

Research Assistant

June 2026 - Present

- Developed and evaluated a foundation model for redshift prediction across millions of astronomical spectra
- Owned and optimized experiment design, benchmarking infrastructure, data preprocessing, and scalable multi-node, multi-GPU training pipelines on NERSC's Perlmutter
- Built agentic workflows to automate experiment execution and evaluation, accelerating hypothesis testing

University of San Francisco, Computer Science Tutor & TA

August 2024 - December 2025

- Quickly debugged and identified issues while explaining complex topics at an elementary level

Sláinte Volleyball Club, Head Coach

September 2021 - July 2025

## PROJECTS

Embedded Systems Arcade Game

Spring 2025 - Present

- Built an ST7796S driver on the Pico SDK for RP2040, achieving 92% of theoretical max fps without visible tearing, despite memory constrained to 2/3 a framebuffer and SPI bandwidth below one frame per refresh
- Brought up distributed Raspberry Pis with custom hardware, GPIO/UART, and real-time input arbitration
- Designed fault-tolerant timing protocols for multi-device sync with 1 controller and many players on 2 lines

Alpha-Zero Style Go Engine

Spring 2026

- Built a self-play pipeline with policy/value CNNs, PUCT MCTS, and player-pool sampling from prior models
- Implemented replay tooling and candidate-vs-champion evaluation gating to compare checkpoints, promote stronger models, and unify data generation, training, and evaluation in a continuous learning loop

Agentic Assistant

Spring 2026

- Architected a multi-user agent runtime as microservices supporting task execution with tool use and multi-step orchestration across planning, retrieval, and actions beyond single-call LLM interactions
- Built personalized RAG memory using vector databases and embedding models to persist user preferences, retrieve long-term context, and ground agent responses and actions in user history

RISC-V Emulator

Spring 2023

- Implemented a RISC-V emulator in C with instruction decoding and execution for basic instruction types, and extended its set-associative cache simulator with multiword block fills and LRU replacement
- Built a five-stage RISC-V processor in Digital with hazard tracking, forwarding, stalls, and pipeline flushing
- Designed a dual-bank memory datapath with interleaved 64-bit banks to support misaligned byte, word, and doubleword loads and stores across memory-word boundaries

Reverse Proxy

Fall 2024

- Led a team of engineers in partnership with Consilient Labs to develop a scalable and efficient reverse proxy as the primary layer of security for a production API with fine-grained permission control

## TECHNICAL SKILLS

**Programming Languages:** C, C++, Python, Rust, JavaScript, Java, RISC-V Assembly

**Systems:** Linux, Embedded Systems, Firmware Bring-Up, GPIO/UART, IPC, Real-Time Constraints, Concurrency, Multithreading, HW/SW Integration, Distributed Systems, Networking, Computer Architecture, Protocols

**Infrastructure:** Git, Docker, AWS, CI/CD, Microservices, Actions, K8s, REST, SQL, noSQL, OAuth, Prometheus/Grafana

**AI/ML:** PyTorch, CNNs, Transformers, Agent Workflows, Agent Infrastructure, LLM Integration, RAG, Self-Play/RL

## COURSEWORK

Operating Systems, Systems Foundations, Compilers, Advanced Algorithms, Machine Learning, Bayesian Deep Learning, Program Synthesis, Advanced Linear Algebra, Cloud Computing